Topic: Solving systems

Question: Which system of differential equations has a coefficient matrix that gives distinct real eigenvalues?

Answer choices:

A $x_1' = 2x_2$ and $x_2' = 4x_2 - 2x_1$

B $x_1' = x_1 + 3x_2$ and $x_2' = x_2 - 3x_1$

C $x_1' = x_1 + 4x_2$ and $x_2' = 2x_1 + 3x_2$

D $x_1' = 6x_1 - 3x_2$ and $x_2' = 3x_1$

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Solution: C

The Eigenvalues of the system in answer choice A are given by

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} -\lambda & 2 \\ -2 & 4-\lambda \end{vmatrix} = 0$$

$$-\lambda(4-\lambda) - 2(-2) = 0$$

$$\lambda^2 - 4\lambda + 4 = 0$$

$$(\lambda - 2)^2 = 0$$

$$\lambda_1 = \lambda_2 = 2$$

The Eigenvalues of the system in answer choice B are given by

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} 1-\lambda & 3 \\ -3 & 1-\lambda \end{vmatrix} = 0$$

$$(1-\lambda)(1-\lambda) - 3(-3) = 0$$

$$(1-\lambda)^2 + 9 = 0$$

$$\lambda_{1,2} = 1 \pm 3i$$

The Eigenvalues of the system in answer choice C are given by

$$|A - \lambda I| = 0$$

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$$\begin{vmatrix} 1-\lambda & 4 \\ 2 & 3-\lambda \end{vmatrix} = 0$$

$$(1-\lambda)(3-\lambda) - 8 = 0$$

$$\lambda^2 - 4\lambda - 5 = 0$$

$$(\lambda + 1)(\lambda - 5) = 0$$

$$\lambda_1 = -1, 5$$

The Eigenvalues of the system in answer choice D are given by

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} 6-\lambda & -3 \\ 3 & 0-\lambda \end{vmatrix} = 0$$

$$(6-\lambda)(0-\lambda) - (-3)(3) = 0$$

$$\lambda^2 - 6\lambda + 9 = 0$$

$$(\lambda - 3)^2 = 0$$

$$\lambda_1 = \lambda_2 = 3$$

The only coefficient matrix that gives distinct real roots is the coefficient matrix for the system in answer choice C.

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Topic: Solving systems

Question: Determine which vector satisfies the system of differential equations.

$$\vec{x}' = \begin{bmatrix} 1 & 2 \\ -1 & 4 \end{bmatrix} \vec{x}$$

Answer choices:

A $\begin{bmatrix} 2 \\ 1 \end{bmatrix} e^{2t}$

B $\begin{bmatrix} -1 \\ 2 \end{bmatrix} e^{-2t}$

C $\begin{bmatrix} 1 \\ 2 \end{bmatrix} e^{2t}$

D $\begin{bmatrix} 2 \\ -1 \end{bmatrix} e^{t}$

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Solution: A

Plugging the vector from answer choice A and its derivative into the matrix equation that represents the system shows that the vector satisfies the system.

$$\begin{bmatrix} 4e^{2t} \\ 2e^{2t} \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ -1 & 4 \end{bmatrix} \begin{bmatrix} 2e^{2t} \\ e^{2t} \end{bmatrix}$$

$$\begin{bmatrix} 4e^{2t} \\ 2e^{2t} \end{bmatrix} = \begin{bmatrix} 2e^{2t} + 2e^{2t} \\ -2e^{2t} + 4e^{2t} \end{bmatrix}$$

$$\begin{bmatrix} 4e^{2t} \\ 2e^{2t} \end{bmatrix} = \begin{bmatrix} 4e^{2t} \\ 2e^{2t} \end{bmatrix}$$

Plugging the vector from answer choice B and its derivative into the matrix equation that represents the system shows that the vector is not a solution to the system.

$$\begin{bmatrix} 2e^{-2t} \\ -4e^{-2t} \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ -1 & 4 \end{bmatrix} \begin{bmatrix} -e^{-2t} \\ 2e^{-2t} \end{bmatrix}$$

$$\begin{bmatrix} 2e^{-2t} \\ -4e^{-2t} \end{bmatrix} = \begin{bmatrix} -e^{-2t} + 4e^{-2t} \\ e^{-2t} + 8e^{-2t} \end{bmatrix}$$

$$\begin{bmatrix} 2e^{-2t} \\ -4e^{-2t} \end{bmatrix} \neq \begin{bmatrix} 3e^{-2t} \\ 9e^{-2t} \end{bmatrix}$$

Plugging the vector from answer choice C and its derivative into the matrix equation that represents the system shows that the vector is not a solution to the system.

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$$\begin{bmatrix} 2e^{2t} \\ 4e^{2t} \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ -1 & 4 \end{bmatrix} \begin{bmatrix} e^{2t} \\ 2e^{2t} \end{bmatrix}$$

$$\begin{bmatrix} 2e^{2t} \\ 4e^{2t} \end{bmatrix} = \begin{bmatrix} e^{2t} + 4e^{2t} \\ -e^{2t} + 8e^{2t} \end{bmatrix}$$

$$\begin{bmatrix} 2e^{2t} \\ 4e^{2t} \end{bmatrix} \neq \begin{bmatrix} 5e^{2t} \\ 7e^{2t} \end{bmatrix}$$

Plugging the vector from answer choice D and its derivative into the matrix equation that represents the system shows that the vector is not a solution to the system.

$$\begin{bmatrix} 2e^{t} \\ -e^{t} \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ -1 & 4 \end{bmatrix} \begin{bmatrix} 2e^{t} \\ -e^{t} \end{bmatrix}$$

$$\begin{bmatrix} 2e^{t} \\ -e^{t} \end{bmatrix} = \begin{bmatrix} 2e^{t} - 2e^{t} \\ -2e^{t} - 4e^{t} \end{bmatrix}$$

$$\begin{bmatrix} 2e^{t} \\ -e^{t} \end{bmatrix} \neq \begin{bmatrix} 0 \\ -6e^{t} \end{bmatrix}$$

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Topic: Solving systems

Question: Which vector set could be a fundamental set of solutions to a system of differential equations?

Answer choices:

A $\begin{bmatrix} -2 \\ 3 \end{bmatrix} e^{-5t}$ and $\begin{bmatrix} 2 \\ 3 \end{bmatrix} e^{7t}$

B $\begin{bmatrix} 1 \\ 2 \end{bmatrix} e^{-5t}$ and $\begin{bmatrix} -2 \\ -4 \end{bmatrix} e^{7t}$

C $\begin{bmatrix} -1 \\ 2 \end{bmatrix} e^{t}$ and $\begin{bmatrix} 3 \\ -6 \end{bmatrix} e^{2t}$

D $\begin{bmatrix} e^{3t} \\ -4e^{3t} \end{bmatrix}$ and $\begin{bmatrix} -2e^{t} \\ 8e^{t} \end{bmatrix}$

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Solution: A

If the Wronskian of a vector set is non-zero, then the vectors are linearly independent and could therefore be a fundamental set of solutions to some system of differential equations.

The Wronskian of the vector set in answer choice A is non-zero.

$$W(\overrightarrow{x}_1, \overrightarrow{x}_2) = \begin{vmatrix} -2e^{-5t} & 2e^{7t} \\ 3e^{-5t} & 3e^{7t} \end{vmatrix}$$

$$W(\overrightarrow{x}_1, \overrightarrow{x}_2) = -2e^{-5t}(3e^{7t}) - 2e^{7t}(3e^{-5t})$$

$$W(\overrightarrow{x}_1, \overrightarrow{x}_2) = -6e^{2t} - 6e^{2t}$$

$$W(\overrightarrow{x}_1, \overrightarrow{x}_2) = -12e^{2t}$$

$$W(\overrightarrow{x}_1, \overrightarrow{x}_2) \neq 0$$

The Wronskian of the vector set in answer choice B is zero.

$$W(\overrightarrow{x}_1, \overrightarrow{x}_2) = \begin{vmatrix} e^{-5t} & -2e^{7t} \\ 2e^{-5t} & -4e^{7t} \end{vmatrix}$$

$$W(\overrightarrow{x}_1, \overrightarrow{x}_2) = -4e^{2t} - (-2e^{7t})(2e^{-5t})$$

$$W(\overrightarrow{x}_1, \overrightarrow{x}_2) = 0$$

The Wronskian of the vector set in answer choice C is zero.

$$W(\overrightarrow{x}_1, \overrightarrow{x}_2) = \begin{vmatrix} -e^{t} & 3e^{2t} \\ 2e^{t} & -6e^{2t} \end{vmatrix}$$

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$$W(\overrightarrow{x}_1, \overrightarrow{x}_2) = 6e^{3t} - 6e^{3t}$$

$$W(\overrightarrow{x}_1, \overrightarrow{x}_2) = 0$$

The Wronskian of the vector set in answer choice D is zero.

$$W(\overrightarrow{x}_1, \overrightarrow{x}_2) = \begin{vmatrix} e^{3t} & -2e^{t} \\ -4e^{3t} & 8e^{t} \end{vmatrix}$$

$$W(\overrightarrow{x}_1, \overrightarrow{x}_2) = 8e^{4t} - 8e^{4t}$$

$$W(\overrightarrow{x}_1, \overrightarrow{x}_2) = 0$$

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